

Residence Radar

Time Series Forecasting Application

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Use Case

When people are looking for a new house ...



**Which
neighborhood
to live in**



**Best
forecasted
house price**



**Potential crime
rate**



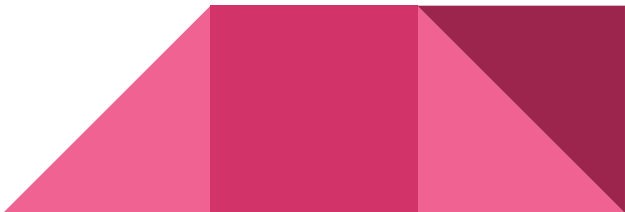
**Income
Demographics**

Jobs to be done

For users/home seekers When I use this app,

1. I want to find out **which neighborhood to live** [Function]
2. I want to know **how expensive the houses** would be in the coming year in my chosen neighborhood/zip code [Function]
3. I want to know the **how safe** in any chosen neighborhood/zip code in the future [Function]
4. I want to learn **how wealthy** in any chosen neighborhood/zip code [Function]
5. I want to find suitable **houses on sale and realtors** in any chosen neighborhood/zip code [Function/Social]

For advertisers/realtors and sellers When I use this app,

1. I want to advertise the house on sale [Function]
 2. I want buyers contact me for business [Function]
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Data

Median income

Median income from 2011
to 2022 in Pittsburgh
by zip code.

Crime rate

Crime rate from 2018 to
2023 in Pittsburgh
by zip code.

House price

Median
House price from 2000 to
2023 in Pittsburgh
by zip code.



Values & Benefits

Value Proposition

Rather than having to scour multiple sources or rely on anecdotal evidence, users can make more informed decisions about where to live based on concrete data.

Product Differentiation

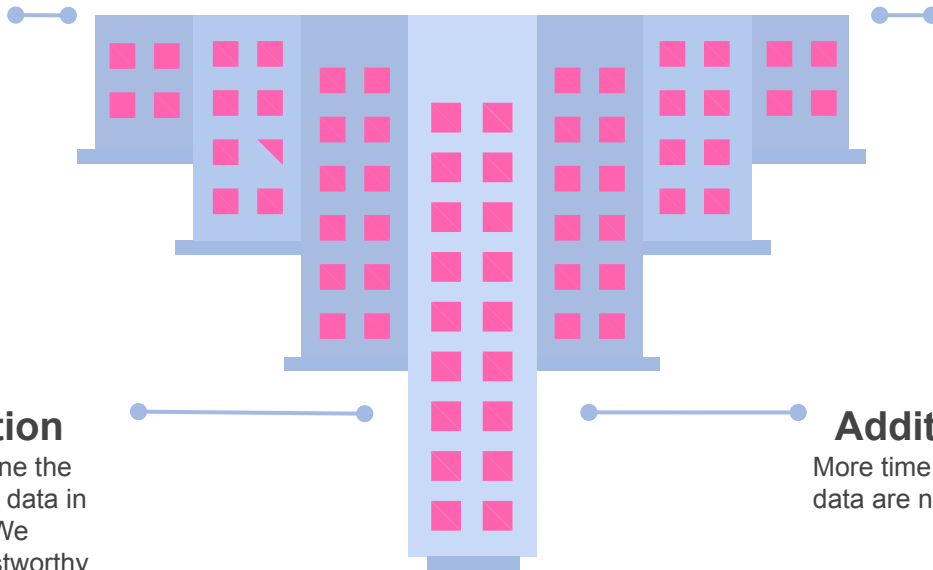
The application aims to streamline the process by providing all essential data in one user-friendly interface. We differentiate the product as a trustworthy and valuable resource.

Economic Benefits

Time savings come from the convenience of having all necessary information in one place. Money savings stem from the ability to make more informed decisions about housing, potentially avoiding costly mistakes such as moving to an area with higher crime rates or overpaying for a property relative to its value.

Additional data for Full Value

More time series data, and safe cloud storage of data are needed to get the full value.



Discovery Hypothesis

CUSTOMER PROBLEM: We believe that individuals seeking housing in Pittsburgh face the challenge of accessing information about specific zip code areas. They need to understand crucial factors such as median income levels, crime rates, and housing prices within each zip code to make informed decisions about where to live.

PROPOSED SOLUTION: We propose to address this problem by developing an intuitive application that allows users to input a specific zip code and instantly access detailed data relevant to that area.

HYPOTHESES UNDERLYING THE IDEA

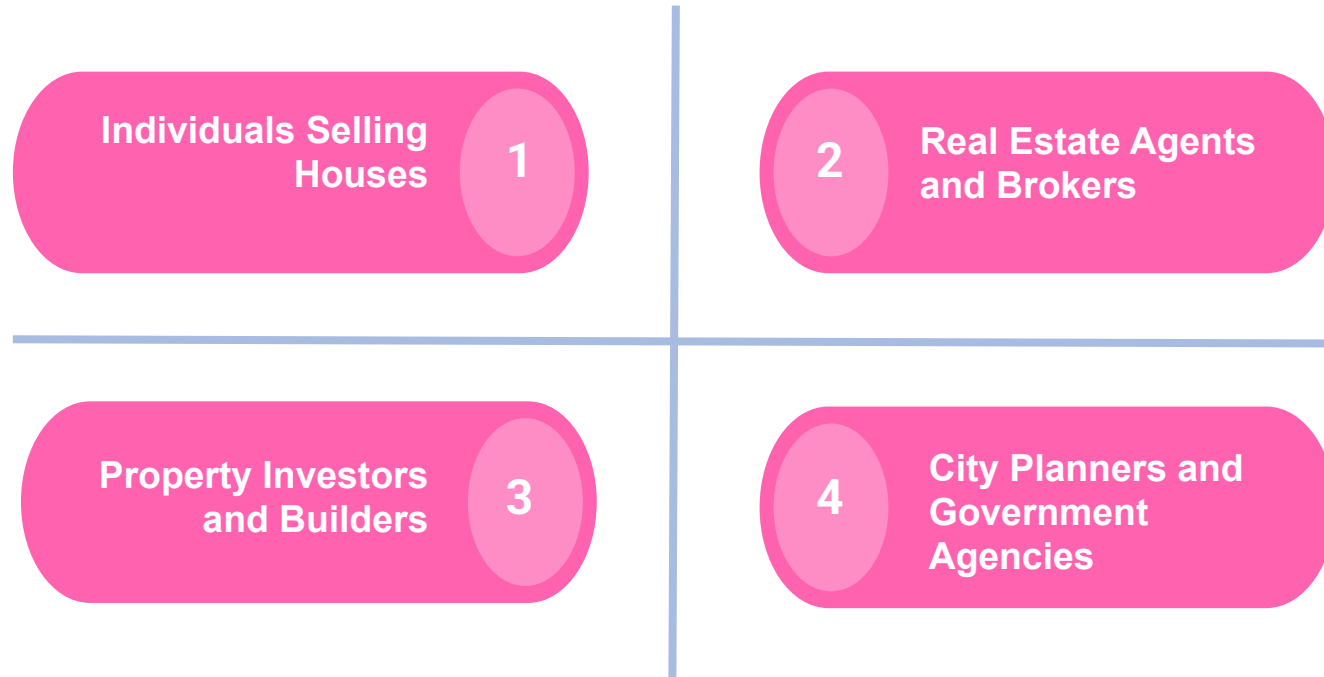
H1: Users will find real-time data about their chosen zip code area valuable, particularly regarding factors such as crime rates and housing prices, enabling them to make timely and informed housing decisions.

H2: Existing approaches to accessing neighborhood data are insufficient, and there is a clear need for a centralized platform that aggregates and presents all relevant data about zip code areas in Pittsburgh to facilitate more effective decision-making.

H3: Users will readily adopt a digital solution for accessing zip code data, recognizing the convenience and utility it offers for housing research purposes despite potential initial hesitations.



Buyers



Buyer Persona - Individual Selling House



SARAH WILLIAM

- Age: 35
- Gender: Female
- Occupation: Marketing Manager
- Income: Moderate
- Location: Pittsburgh, PA
- Family: Married with two children

GOALS

- Seeking to sell a property in Pittsburgh.
- Interested in understanding the local housing market to set competitive prices.

PERSONALITY

- Detail-oriented and organized
- Pragmatic and resourceful
- Values transparency and honesty in business dealings

PAIN POINTS

- Difficulty in accurately pricing their property based on market trends and comparable listings.
- Concerns about finding reliable tenants or buyers who meet their criteria.
- Desire to showcase neighborhood features and amenities to attract potential buyers.

OBJECTIONS

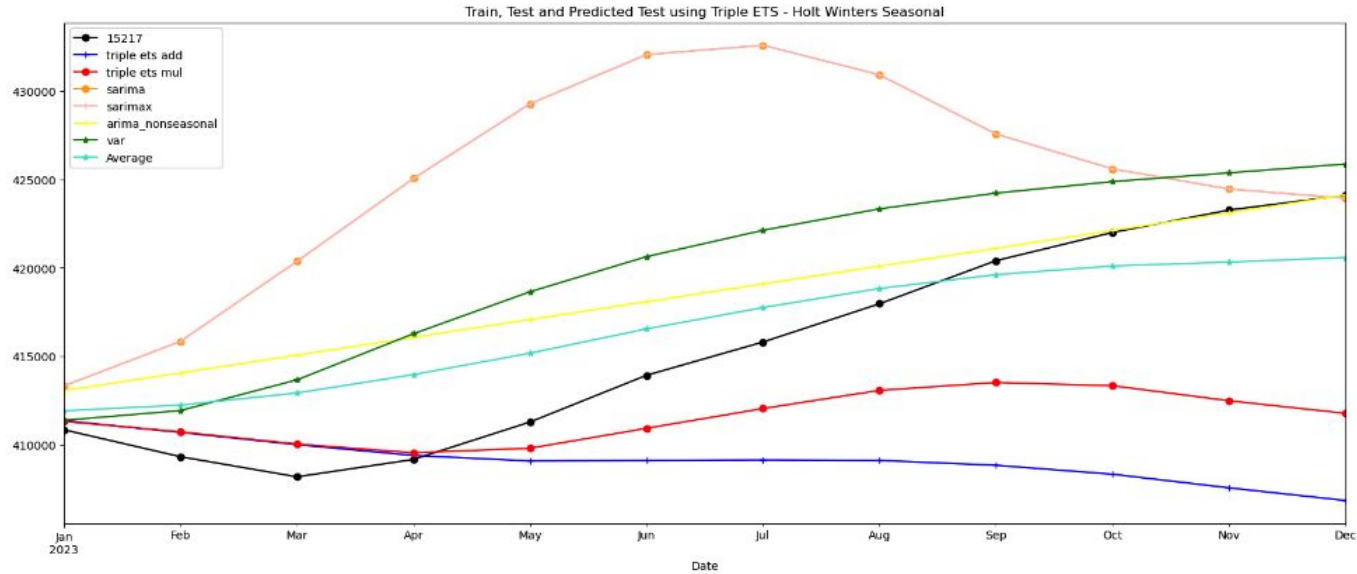
- Skepticism about the effectiveness of traditional real estate agents.
- Fear of making costly mistakes or being taken advantage of by buyers.
- Reluctance to negotiate or compromise on the sale price or terms of the agreement.

Forecasting Techniques

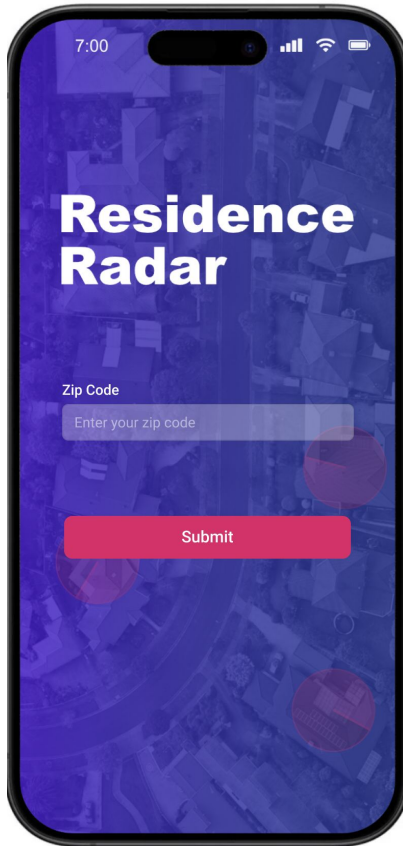
- Applying advanced Times Series Forecasting models
- Extensive Training Period for Machine Learning
- Backtesting Across Four Distinct Test Periods
- Utilizing Cross-Validation to Optimize Model Fit
- Averaging the Performances of the Top Three Models

	Model	RMSE	MAE	MAPE
6	average	2986.297272	2667.386514	0.643876
5	arma_nonseasonal	3970.632257	3090.750235	0.751370
3	VAR	4884.183125	4339.767079	1.047581
1	Triple Mul	6098.569350	4661.474614	1.109823
0	Triple Add	9230.699235	7059.110328	1.680122
2	SARIMA	11611.050149	9589.291169	2.319757
4	SARIMAX	11611.051028	9589.292325	2.319758

Plot of Predictions vs Actuals



The black line is the actuals. Our forecast models are in turquoise.



WireFrame Demo

Questions?

